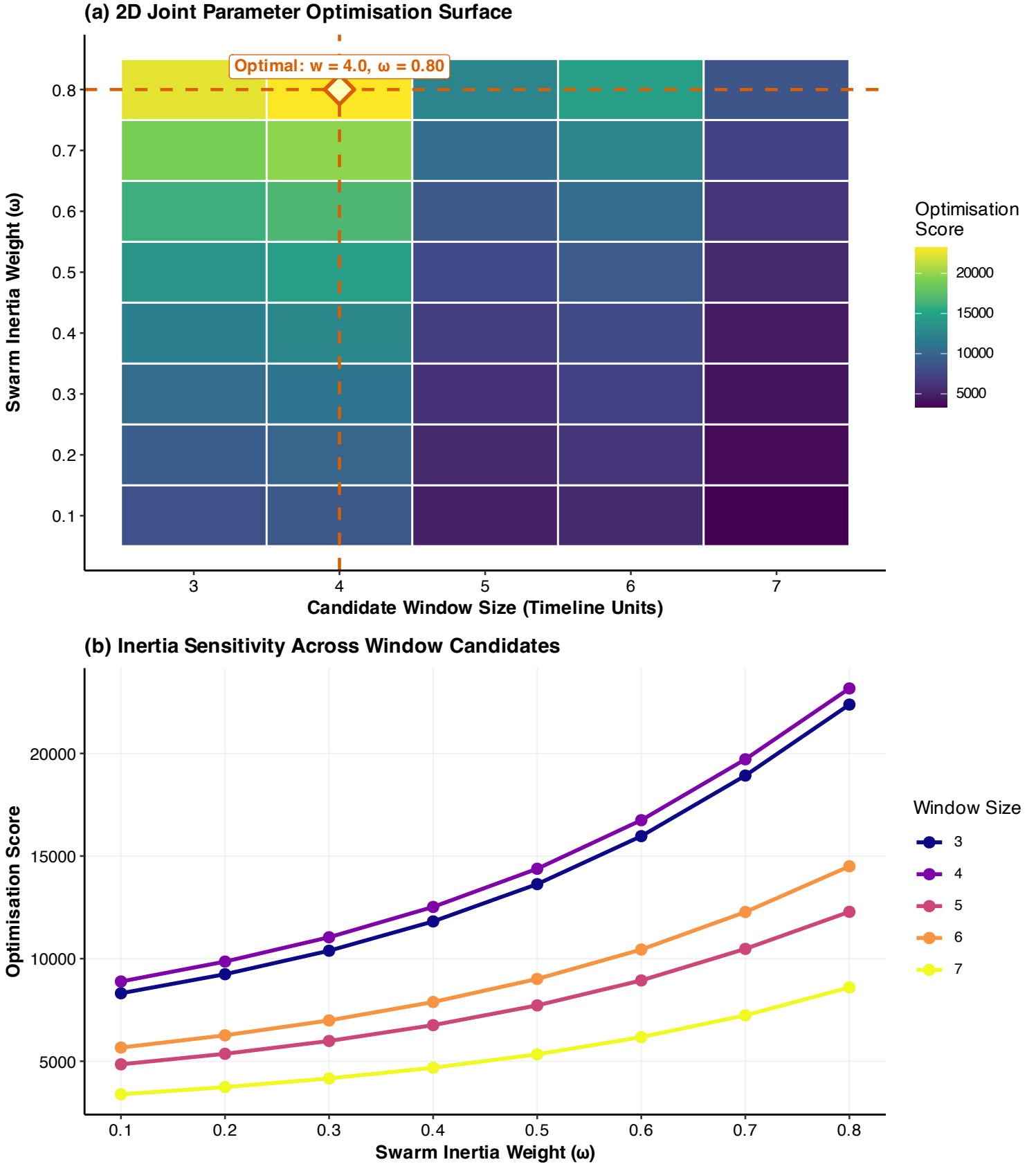
(a) 2D Joint Parameter Optimisation Surface
Optimal: w = 4.0, ω = 0.80
Swarm Inertia Weight (ω)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
3
4
5
6
7
Candidate Window Size (Timeline Units)
Optimisation Score
20000
15000
10000
5000
(b) Inertia Sensitivity Across Window Candidates
Optimisation Score
20000
15000
10000
5000
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Swarm Inertia Weight (ω)
Window Size
3
4
5
6
7